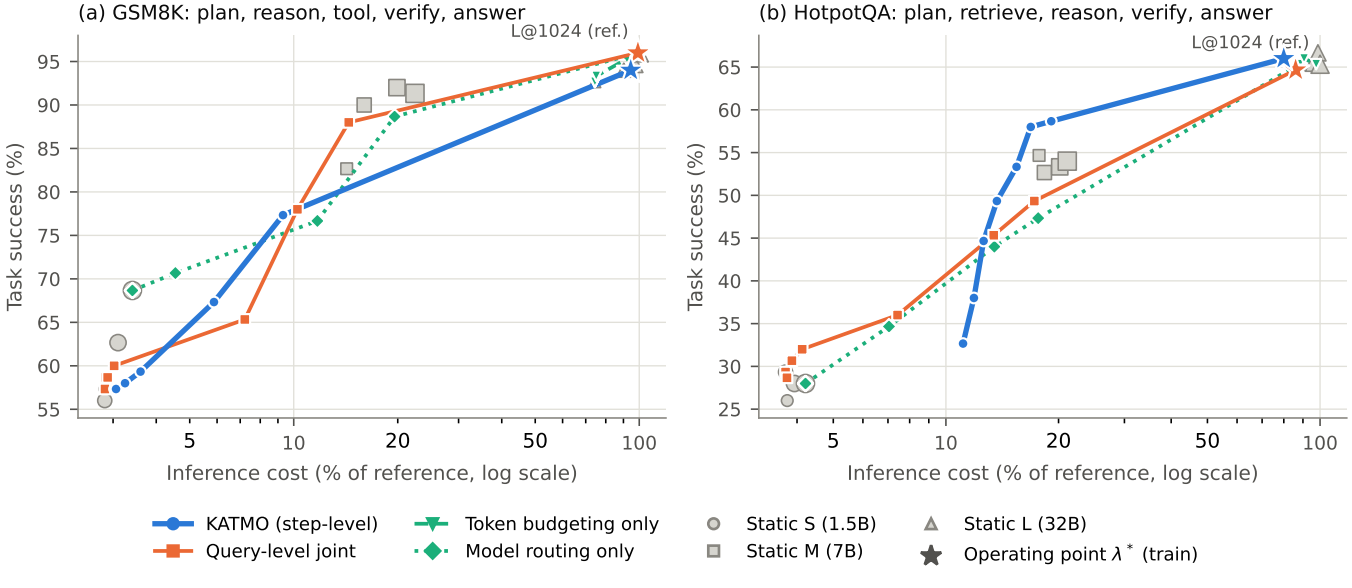
(a) GSM8K: plan, reason, tool, verify, answer
L@1024 (ref.)
Task success (%)
95
90
85
80
75
70
65
60
55
5
10
20
50
100
Inference cost (% of reference, log scale)
(b) HotpotQA: plan, retrieve, reason, verify, answer
L@1024 (ref.)
Task success (%)
65
60
55
50
45
40
35
30
25
5
10
20
50
100
Inference cost (% of reference, log scale)
KATMO (step-level)
Query-level joint
Token budgeting only
Model routing only
Static S (1.5B)
Static M (7B)
Static L (32B)
Operating point λ* (train)